Zihan Guo

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EDUCATION

University of Toronto **Toronto, CA**

B.S in Data Science Specialist & Statistics Major,Economics Minor(Honor Program) Sep.2023 – Apr.2027

Core Courses: Software Design, Data Structures & Analysis, Data Science II, Intro Machine Learning, Time Series Analysis, Neural Nets and Deep Learning, Algorithm Design & Analysis, Data Visualization

PUBLICATION & MANUSCRIPTS

- **Z. Guo,** R. He, J. Quan. Before Bringing It Up: When and How AI Companions Should Use Memory, ACM CHI Conference on Human Factors in Computing Systems, 2027 - Under Review.
- J. Huang, J. Quan, S. Lyu, **Z. Guo,** Y. Ge. SkillScreen: High-Recall Screening of Malicious Agent Skills with Decision Models, *NAACL, 2027* - Submitted.

RESEARCH EXPERIENCE

SkillScreen: High-Recall Screening of Malicious Agent Skills with Decision Models **Sep.2026 - Present**

- *NAACL 2027 — Under Review*

- Developed a 2B-parameter natural language inference (NLI) model for single-pass malicious agent skill screening; applied LoRA fine-tuning with 2.19M trainable parameters to enable efficient classification.
- Benchmarked the model across five random seeds on MaliciousSkillBench's 1,947-skill test set, including unseen attack families, achieving 95.6% recall, a 3.3% false-positive rate, and 0.992 AUROC.
- Optimized screening thresholds to maintain 99% validation recall, attaining 98.7–99.1% test recall at 20.1–23.9% false-positive rates; assessed unseen-source generalization, identifying unfamiliar benign sources as a key deployment challenge.

Before Bringing It Up: Memory-Aware Interaction Design for AI Companions **Jul.2026 -Present**

- *CHI 2027 — Under Review, Co-first Author*

- Identified design requirements for memory use in AI companionship through two rounds of formative interviews with 14 users; synthesized eight design themes from the memory-focused round to guide interaction design.
- Developed Reconsider, a single-call prompting procedure with five checks and four handling modes to assess memory validity, permission, contextual value, timing, and scope before AI companions retrieve stored memories.
- Evaluated 800 responses across five LLMs and 80 scenarios using 400 blinded comparisons; two LLM judges reported +15.3 and +22.8 percentage-point net win margins over baseline.

University of Toronto | Dynamic Graphics Project Lab **Oct.2026 – Present**

Undergraduate Researcher with Prof. Khai N. Truong

- Reviewed AI-enabled social interaction systems for dementia care, categorizing applications by agent embodiment, interaction roles, and care contexts while comparing outcomes in user acceptance and effectiveness.
- Conduct design and prototyping work for companion AI agents, exploring how design elements such as conversational styles, personalized memory, and interaction pacing influence the user experience.
- The evaluation will focus on the system's usability, cognitive load, companionship experience, and trust, while also considering privacy, user autonomy, and the boundaries of caregiver involvement.

University of Toronto | Guttman Laboratory of Pathogen Genomics & Evolution **May.2025 - May.2026**

Research Assistant with Prof. David S. Guttman

- Developed a reproducible R workflow for bacterial insertion-sequence (IS) analysis, standardizing genome annotations and curating 48 candidate records from 141 ISEscan predictions across two genome assemblies.
- Validated 61 of 65 IS predictions (93.8%) against ISfinder/BLAST hits using genomic coordinates, resolving discrepancies through sequence and flanking-region comparisons to distinguish IS21 copies from duplicates.
- Characterized strain-level insertion-sequence patterns across bacterial genomes through R-based analysis and visualization, translating genomic comparisons into biological interpretations for lab discussions.

SELECTED PROJECTS

Echoloom | CSC454 *Project* **Jul - Sep. 2026**

- Analyzed competitor products and 6 user interviews to identify use cases, willingness-to-pay signals, and experience gaps, shaping target-user definition and product positioning for an AI emotional-companion MVP.
- Translated user insights into a story-first UX centered on sustained AI relationships, defining a 5-character experience and prioritizing MVP features by user value, development cost, and technical feasibility.
- Built a Gemini-powered Node.js back-end that maintained character-specific context and validated AI outputs,

improving dialogue continuity through a persistent memory architecture with recovery and fallback mechanisms.
- Developed a bilingual HTML/CSS/JavaScript prototype covering the core narrative and interaction flow, with 5 investigation chapters and 3 endings; validated the implementation through 6 automated test suites.

Tracing the AI Narrative: What The Guardian's Coverage Reveals | JSC370 *Project* **Feb - Apr.2026**
- Constructed a corpus of 40,232 Guardian articles via Python/Guardian API, filtering 3,120 AI-related records with keyword and headline/summary matching; validated the filter through 200 manual reviews.
- Developed an scikit-learn classification pipeline using TF-IDF and metadata, with XGBoost achieving 76.44% accuracy, 0.750 F1, and 0.847 ROC-AUC, versus 53.37% for the majority baseline.
- Applied LDA to identify 6 topics and track monthly shifts in AI coverage; visualized results with Plotly/Quarto and documented modeling assumptions and data limitations.

ChartCreator| Personal Project **Sep.2025 - Mar.2026**
- Designed an AI-powered visualization platform that mapped data set structure to context-aware chart recommendations, reducing chart creation to a 3-step process for non-technical users.
- Built multi-turn prompts and recommendation logic to classify numerical, categorical, and time-series variables and map them to 8 common chart types; validated performance on 30+ datasets with an 85%+ manual review pass rate.
- Developed a JavaScript interactive prototype with 6 core modules and 15+ UI states, reducing basic chart creation time from 10 – 15 minutes to 1 – 3 minutes.

CourseInsight: University of Toronto Course Finder | Personal Project **Jan.2026 - Present**
- Built a reproducible Python/pandas pipeline integrating 10,037 official courses and 7,093 offering records across three campuses into a 10-table DuckDB database, preserving source links and retrieval dates.
- Integrated 70,453 community-archived historical survey reports, exactly matching 54,196 reports to 6,505 current courses while preserving original rating scales, source attribution, and unmatched records.
- Implemented coverage checks, SHA-256 verification, and atomic database publication to prevent incomplete or invalid updates from overwriting verified snapshots.
- Developed a Streamlit app with campus, level, subject, and breadth/distribution filters, offering status, course comparison, CSV export, and access to 752 indexed learning resources.

PROFESSIONAL EXPERIENCE

Zhongke Brain Intelligence Technology Co., Ltd. | *Algorithm Intern* **May - Jul.2025 / Hefei, CN**
- Processed 50,000+ power-generation records with 20+ raw features, integrating operational, meteorological, and temporal data to construct multivariate time-series forecasting inputs.
- Built 30+ temporal and operational features and generated supervised samples through sliding-window construction, preparing sequential inputs for LSTM/GRU forecasting models.
- Evaluated forecasting performance using MAE, RMSE, and MAPE, conducting error analysis to identify high-deviation periods and inform monitoring-rule refinement and subsequent model iterations.

SKILLS

Programming Languages: Python | SQL | R | Java | JavaScript | Bash
Software & Tools: MySQL | DuckDB | Pandas | Excel | Tableau | Streamlit | Plotly | R | Git | Docker | Linux
Certification: Oracle MySQL 8.0 Database Developer Certified Professional | Oracle Cloud Infrastructure 2025 Certified Foundations Associate